

Shaopeng Fu

shaopeng.fu@kaust.edu.sa | +966 56 534 0337 | <https://shaopengfu.me>

Google Scholar: [i7cWm4gAAAAJ](#) | LinkedIn: [shaopeng-fu-762498194](#) | GitHub: [fshp971](#)

SUMMARY

Ph.D. candidate at KAUST and research intern at Microsoft Research Asia working on **reliable and efficient LLM post-training, adversarial robustness, and scalable evaluation**. I combine deep learning theory with system-level optimization to improve the training and evaluation of reasoning models. Recent work spans RL post-training for coding LLMs, LLM adversarial training, and adversarial-training-inspired data synthesis.

EDUCATION

King Abdullah University of Science and Technology Aug. 2023 – Present (Expected Jun. 2027)
Ph.D. Candidate in Computer Science (Advisor: Prof. Di Wang) Thuwal, Saudi Arabia
Topics: LLM adversarial robustness, deep learning theory

The University of Sydney Oct. 2019 – Jan. 2021
Master of Philosophy (Engineering and IT) (Advisor: Prof. Dacheng Tao) Sydney, Australia
Thesis: “[Bayesian Inference Forgetting](#)”

South China University of Technology Sep. 2015 – Jun. 2019
B.Sc. in Mathematics and Applied Mathematics (Advisor: Prof. Chuhua Xian) Guangzhou, China

RESEARCH & INDUSTRY EXPERIENCE

Microsoft Research Asia (MSRA) Jun. 2025 – Present
Research Intern, General Artificial Intelligence ([GenAI](#)) Group (Mentor: Dr. Xingxing Zhang) Beijing, China (Hybrid)
Topic: LLM code generation for competitive programming (CP) and software engineering (SWE)

- **RefineRL for CP:** Developed an RL method (first-author paper under review) to instill self-refinement in Qwen3-4B-2507, trained for 4 days on 32 A100 GPUs. Achieved **75.97% (v5) and 67.82% (v6) pass@1** on LiveCodeBench, outperforming leading 32B models and approaching the performance of Qwen3-235B-A22B-2507.
- **CP evaluation infra:** Architected a concurrent, extensible multi-language pipeline for LLM-generated code using Python asynchronous subprocesses. Reduced evaluation time for 200+ solutions from **>1 hour to <3 minutes** (>20x speedup) versus HF’s Liteval on a 20+ core CPU node, and integrated it with Verl/Ray for distributed RL training.
- **Adversarial SWE-agent RL:** Developing an end-to-end agentic RL framework for SWE environment building and issue resolution, with test-script verifiers adversarially generated during training to reduce reliance on prebuilt environments and manually constructed verifiers.

JD Explore Academy, JD.com, Inc. Mar. 2021 – Jul. 2022
Algorithm Engineer, Trustworthy AI Research Group (Leader: Prof. Dacheng Tao) Beijing, China

- Led development of **TAICore**, a trustworthy AI assessment toolkit for evaluating the robustness and privacy-preserving properties of white-box and black-box ML models.
- Published **two first-author ICLR 2022 papers** on privacy in machine learning.

PUBLICATIONS

* indicates equal contribution.

Selected Papers

1. [Preprint] Shaopeng Fu, Xingxing Zhang, Li Dong, Di Wang, and Furu Wei. “[RefineRL: Advancing Competitive Programming with Self-Refinement Reinforcement Learning](#).” *arXiv preprint arXiv:2604.00790*, 2026.
2. [Preprint] Hongli Shen*, Shaopeng Fu*, Qinbo Zhang, Jian Li, and Di Wang. “[Dual-Adversarial Safety Alignment: Cultivating Intrinsic Threat Comprehension in LLMs](#).” *arXiv preprint arXiv:2608.09542*, 2026. [Code]
3. [Preprint] Xinhai Wang*, Shaopeng Fu*, Shu Yang, Liangyu Wang, Tianhang Zheng, and Di Wang. “[Accelerating Suffix Jailbreak Attacks with Prefix-Shared KV-Cache](#).” *arXiv preprint arXiv:2603.13420*, 2026. [Code]
4. [ICLR’26] Shaopeng Fu and Di Wang. “[Understanding and Improving Continuous LLM Adversarial Training via In-Context Learning Theory](#).” *International Conference on Learning Representations*, 2026. [Code]
5. [NeurIPS’25] Shaopeng Fu, Liang Ding, Jingfeng Zhang, and Di Wang. “[Short-length Adversarial Training Helps LLMs Defend Long-length Jailbreak Attacks: Theoretical and Empirical Evidence](#).” *Advances in Neural Information Processing Systems*, 2025. [Code]
6. [Preprint] Shaopeng Fu, Xuexue Sun, Ke Qing, Tianhang Zheng, and Di Wang. “[Pre-trained Encoder Inference: Revealing Upstream Encoders in Downstream Machine Learning Services](#).” *arXiv preprint arXiv:2408.02814*, 2024. [Code]

7. [ICLR'24 & IEEE TIT] **Shaopeng Fu** and Di Wang. “Theoretical Analysis of Robust Overfitting for Wide DNNs: An NTK Approach.” *International Conference on Learning Representations*, 2024. [Code]
(Also in *IEEE Transactions on Information Theory* 71(8), 2025.)
8. [ICLR'22] **Shaopeng Fu**, Fengxiang He, Yang Liu, Li Shen, and Dacheng Tao. “Robust Unlearnable Examples: Protecting Data Against Adversarial Learning.” *International Conference on Learning Representations*, 2022. [Code]
9. [ICLR'22] **Shaopeng Fu***, Fengxiang He*, and Dacheng Tao. “Knowledge Removal in Sampling-based Bayesian Inference.” *International Conference on Learning Representations*, 2022. [Code]

Other Papers

10. [ICML'26] Jiaming Zhang, Meng Ding, **Shaopeng Fu**, Jingfeng Zhang, and Di Wang. “Benign Overfitting in Adversarial Training for Vision Transformers.” *Forty-Third International Conference on Machine Learning*, 2026.
11. [ICML'26] Meng Ding, Mingxi Lei, **Shaopeng Fu**, Shaowei Wang, Di Wang, and Jinhui Xu. “Understanding Private Learning from Feature Perspective.” *Forty-Third International Conference on Machine Learning*, 2026.
12. [Preprint] Jiaming Zhang, Huanyi Xie, Meng Ding, **Shaopeng Fu**, Jinyan Liu, and Di Wang. “Understanding the Impact of Differentially Private Training on Memorization of Long-Tailed Data.” *arXiv preprint arXiv:2602.03872*, 2026.
13. [Neural Computation] Zeke Xie, Fengxiang He, **Shaopeng Fu**, Issei Sato, Dacheng Tao, and Masashi Sugiyama. “Artificial Neural Variability for Deep Learning: On Overfitting, Noise Memorization, and Catastrophic Forgetting.” *Neural Computation* 33(8), 2021. [Code]
14. [Preprint] Fengxiang He*, **Shaopeng Fu***, Bohan Wang*, and Dacheng Tao. “Robustness, Privacy, and Generalization of Adversarial Training.” *arXiv preprint arXiv:2012.13573*, 2020. [Code]

SELECTED AWARDS

International Collegiate Programming Contest (ICPC)

- **Gold Medal**: Asia Regional Contest Shenyang Site (2018; **Rank: 6/186**)
- **Silver Medals (3x)**: Asia-East Continent Final Xi'an Site (2018), Asia Regional Contest Qingdao Site (2017), and Asia Regional Contest Xi'an Site (2017)

China National Scholarship (2x): 2017 & 2018

ICML Silver Reviewer: 2026

ACADEMIC SERVICE

Conference Reviewer: ICML (2022 – 2026), ICLR (2022 – 2026), NeurIPS (2021 – 2026), AISTATS (2021, 2024 – 2026).

Conference Committee Member: ACM CCS (2024 Artifact Evaluation), AAAI (2025).

Journal Reviewer: Neurocomputing, TMLR, IEEE TIT, IEEE TPAMI, IEEE TNNLS, IEEE TCYB, Springer NPL.

TEACHING

Teaching Assistant, CS 226: Trustworthy Machine Learning	Spring 2026, KAUST
Teaching Assistant, CS 229: Machine Learning	Spring 2024, KAUST

TECHNICAL SKILLS & MISCELLANEOUS

Competitive Programming: Codeforces **2100+** (Master); Profile: [fshp971](#).

Programming Languages: C/C++, Python, Golang

Frameworks & Tools: PyTorch, JAX, vLLM, SGLang, Verl, Docker, Slurm, Linux, Vim